

KRISHNA DUBEY

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Professional Summary

Computer Science undergraduate specializing in AI, Machine Learning, Generative AI, and Full-Stack Development. Proven track record in building scalable AI pipelines, NLP systems, and cloud-native applications. Smart India Hackathon 2025 Grand Finalist with research experience from IIT (BHU), Varanasi focusing on energy-efficient routing and contextual AI models.

Technical Skills

Languages & Web: Python, Java, JavaScript, SQL, HTML5, CSS3

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers, LLMs, NLP

Backend & Full-Stack: FastAPI, React.js, REST APIs, JWT Authentication, Database Design, Docker, Redis

Data & Tools: Pandas, NumPy, Git/GitHub, Linux, Streamlit, Gradio, Jupyter Notebook, CI/CD

Networking (WSN): Energy-Efficient Routing (LEACH, PEGASIS), AODV, DSR routing protocols

Experience

Indian Institute of Technology (BHU), Varanasi – *Summer Research Intern*

May '26 – Jul '26

- Conducted research on WSN architectures, evaluating energy-efficient routing protocols (LEACH, PEGASIS, AODV, DSR).
- Developed and benchmarked deep learning models for sequence processing and optimized feature extraction pipelines.
- Collaborated on building scalable AI microservices utilizing Python, FastAPI, React.js, and secure JWT authentication.

IBM Developer Skills Network – *Generative AI Intern*

Jul '25 – Aug '25

- Programmed 3+ Generative AI applications leveraging OpenAI/Hugging Face APIs and orchestrated low-latency pipelines.
- Implemented few-shot prompting strategies, improving contextual accuracy and response relevance by 25%.
- Designed interactive application frontends using Gradio, deploying containerized workflows for cross-team evaluation.

Social Chammach – *Business Development Associate Intern*

Jul '25 – Sep '25

- Managed outbound marketing operations, optimizing client acquisition pipelines and presenting technology integration solutions.
- Coordinated cross-functional teams to streamline digital campaign deliveries, improving active lead engagement metrics.

Projects

PersonaAI – Personality Prediction & Speech Emotion Recognition

2026

- Engineered a full-stack platform integrating Big Five Personality Prediction and Speech Emotion Recognition (SER), achieving **89% classification accuracy** using PyTorch, Scikit-learn, and Librosa (**MFCCs, chroma extraction**).
- Built an optimization pipeline utilizing TF-IDF, contextual embeddings, and fine-tuned BERT models to process **10k+ character text inputs** in under **1.2 seconds**.
- Architected scalable REST APIs via FastAPI with JWT, reducing query latency by **25%** using optimized SQL indexes.

Audio Language Model (SIH 2025 Grand Finalist)

2025

- Developed an end-to-end, real-time speech-to-text and intent-recognition system using PyTorch, achieving an **8.5% Word Error Rate (WER)**.
- Designed robust raw-signal processing pipelines, integrating model quantization that reduced CPU inference latency by **40%**.
- Selected as a national **Grand Finalist at Smart India Hackathon (SIH) 2025** out of over **50,000+ team submissions**.

Sociogram – Full-Stack Social Media Web Application

2025

- Developed a multi-tier social application using React.js, FastAPI, and PostgreSQL, securing user endpoints with **OAuth 2.0 and JWT**.
- Integrated a **Redis-backed caching layer** and optimized connection pooling, lowering page load times by **45%** under peak loads.
- Containerized microservices using **Docker** and established automated **CI/CD workflows** via GitHub Actions.

AI Social Media Caption Generator

2025

- Built an automated, AI-powered caption generator using OpenAI GPT APIs and Gradio, improving copywriting speeds by **10x**.
- Applied advanced prompt engineering and few-shot learning to elevate brand alignment and text relevance scores by **35%**.
- Integrated NLTK-driven sentiment analysis filters to ensure automatic content validation and brand safety guidelines.

Leadership & Key Achievements

Leadership & Initiatives

- Event & Op Head:** GeeksforGeeks Chapter, BIT: Directed outreach strategies, leading 5+ hackathons for 500+ students.
- Social Media Head** – E-Cell, BIT: Managed digital campaign pipelines, boosting campus event registration by 40%.

Competitive Hackathons & Certifications

- Smart India Hackathon** – Grand Finalist: Designed real-time edge speech models, presenting to national jury. *Dec '25*
- TechSprint (GDG Collaboration)** – 2nd Position: Led rapid-deployment full-stack project cycle in a sprint. *2025*
- IBM Generative AI Professional Certification** | Finalist: Hacknovate 7.0 | Selected: AICTE IDE Bootcamp Jodhpur

Education

Buddha Institute of Technology, Gorakhpur (AKTU)

Expected 2027

Bachelor of Technology in Computer Science and Engineering (AI & ML Specialization)

CGPA: 7.5/10

Relevant Coursework: Data Structures, Object-Oriented Programming, DBMS, Operating Systems, Software Engineering.